

Lab 3

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.1
## Warning: package 'ggplot2' was built under R version 4.5.2
## Warning: package 'tidyr' was built under R version 4.5.1
## Warning: package 'readr' was built under R version 4.5.1
## Warning: package 'purrr' was built under R version 4.5.1
## Warning: package 'dplyr' was built under R version 4.5.1
## Warning: package 'stringr' was built under R version 4.5.1
## Warning: package 'forcats' was built under R version 4.5.1
## Warning: package 'lubridate' was built under R version 4.5.1
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    4.0.2      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(plotrix)
```

```
## Warning: package 'plotrix' was built under R version 4.5.2
```

```
density <- read_csv("../data_raw/Lab3_Density.csv")
```

```
## Rows: 20 Columns: 7
## -- Column specification -----
## Delimiter: ","
## dbl (7): quadrat_number, 50x50cm, 1x1m, 2.5x2.5m, 5x5m, 10x10m, distance_nea...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
time_density <- read_csv("../data_raw/Lab3_densitytime.csv")
```

```
## Rows: 4 Columns: 7
## -- Column specification -----
## Delimiter: ","
## dbl (7): quadrat_number, 50x50cm, 1x1m, 2.5x2.5m, 5x5m, 10x10m, distance_nea...
##
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
time_frequency <- read_csv("../data_raw/Lab3_frequency_time.csv")

## Rows: 4 Columns: 7
## -- Column specification -----
## Delimiter: ","
## dbl (7): quadrat_number, 50x50cm, 1x1m, 2.5x2.5m, 5x5m, 10x10m, distance_nea...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

1.) Calculate the mean and standard error of mesquite density for each plot size and make a graph of the comparison. (Be sure to convert density)

```
#longer data tables
density_data <- density %>%
  pivot_longer(
    cols = c(`50x50cm`, `1x1m`, `2.5x2.5m`, `5x5m`, `10x10m`),
    names_to = "plot_size",
    values_to = "count"
  ) %>%
  mutate(
    # Set the area in m^2 for each plot size
    area_m2 = case_when(
      plot_size == "50x50cm" ~ 0.25,
      plot_size == "1x1m" ~ 1.00,
      plot_size == "2.5x2.5m" ~ 6.25,
      plot_size == "5x5m" ~ 25.00,
      plot_size == "10x10m" ~ 100.00),
    #convert density
    density = count/area_m2,
    #set as factor to order smallest to largest quadrat and order x-axis titles
    plot_size = factor(plot_size,
      levels = c("50x50cm", "1x1m", "2.5x2.5m", "5x5m", "10x10m")
    )
  )

#standard error and standard deviation
summary_density <- density_data %>%
  group_by(plot_size) %>%
  summarise(
    mean_density = mean(density),
    se_density = sd(density) / sqrt(n())
  )

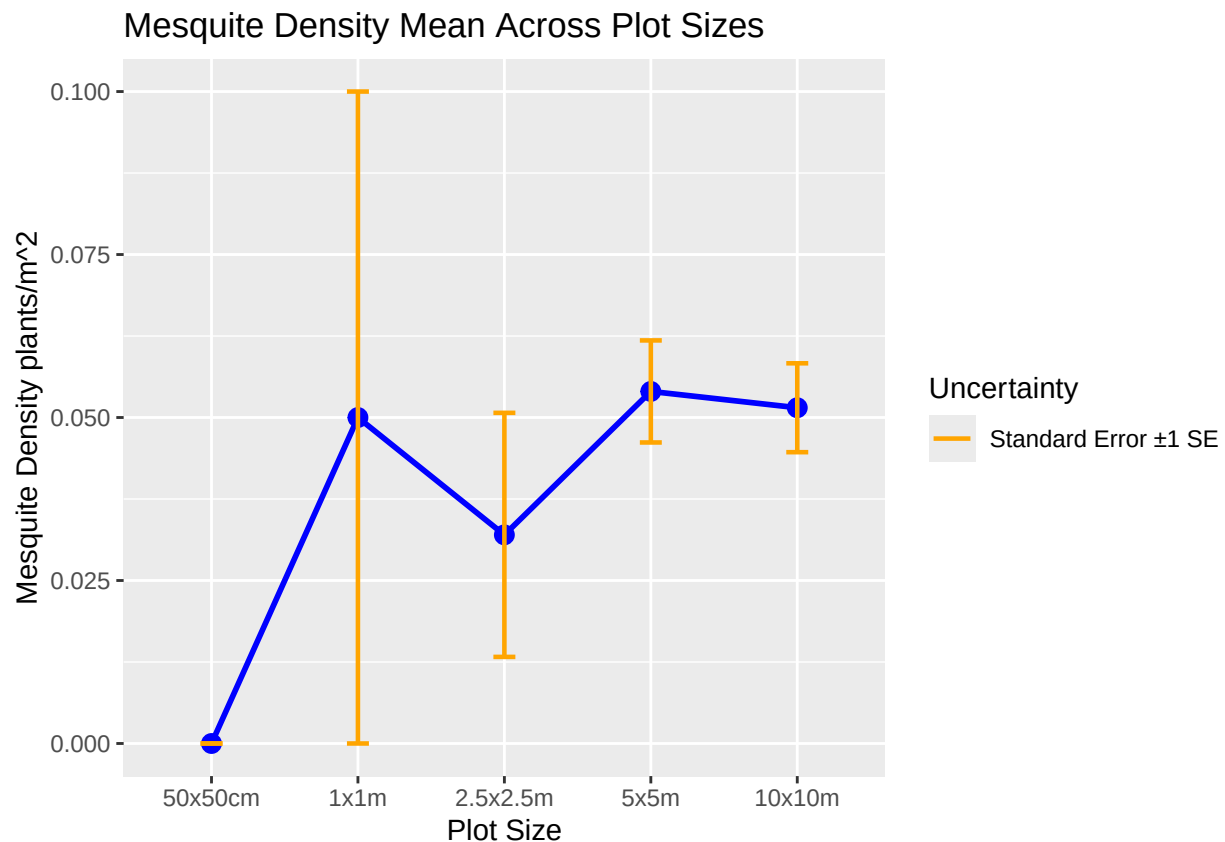
print(summary_density)

## # A tibble: 5 x 3
##   plot_size mean_density se_density
##   <fct>      <dbl>      <dbl>
## 1 50x50cm      0          0
```

```
## 2 1x1m          0.05      0.05
## 3 2.5x2.5m      0.032     0.0187
## 4 5x5m          0.054     0.00783
## 5 10x10m        0.0515    0.00682
```

```
#plot summary stats
ggplot(summary_density, aes(x = plot_size, y = mean_density, group = 1)) +
  geom_line(color = "blue", size = 1) +
  geom_point(color = "blue", size = 3) +
  geom_errorbar(
    aes(ymin = mean_density - se_density, ymax = mean_density + se_density,
        color = "Standard Error ±1 SE"),
    width = 0.15, linewidth = 0.8
  ) +
  scale_color_manual(name = "Uncertainty", values = c("Standard Error ±1 SE" = "orange")) +
  labs(
    title = "Mesquite Density Mean Across Plot Sizes",
    x = "Plot Size",
    y = "Mesquite Density plants/m^2")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



2.) Calculate the average time to sample each plot size.

```
time_density_longer <- time_density %>%
  pivot_longer(
    cols = c(`50x50cm`, `1x1m`, `2.5x2.5m`, `5x5m`, `10x10m`),
    names_to = "plot_size",
    values_to = "count")

time_density_longer <- time_density_longer %>%
  mutate(
    plot_size = factor(
      plot_size,
      levels = c("50x50cm", "1x1m", "2.5x2.5m", "5x5m", "10x10m") ) )

summary_time_density <- time_density_longer %>%
  group_by(plot_size) %>%
  summarise(mean_time = mean(count))

cat("Mean sampling time for density in seconds")

## Mean sampling time for density in seconds
print(summary_time_density)
```

```
## # A tibble: 5 x 2
##   plot_size mean_time
##   <fct>      <dbl>
## 1 50x50cm      7.75
## 2 1x1m        5.75
## 3 2.5x2.5m    15.2
## 4 5x5m       16.2
## 5 10x10m     28
```

```
time_frequency_longer <- time_frequency %>%
  pivot_longer(
    cols = c(`50x50cm`, `1x1m`, `2.5x2.5m`, `5x5m`, `10x10m`),
    names_to = "plot_size",
    values_to = c("count"))

time_frequency_longer <- time_frequency_longer %>%
  mutate(
    plot_size = factor( plot_size,
      levels = c("50x50cm", "1x1m", "2.5x2.5m", "5x5m", "10x10m")
    ) )

summary_time_frequency <- time_frequency_longer %>%
  group_by(plot_size) %>%
  summarise(mean_time = mean(count))

cat("Mean sampling time for frequency in seconds")

## Mean sampling time for frequency in seconds
print(summary_time_frequency)
```

```
## # A tibble: 5 x 2
```

```
##   plot_size mean_time
##   <fct>      <dbl>
## 1 50x50cm    2.25
## 2 1x1m       4.25
## 3 2.5x2.5m   7.5
## 4 5x5m       7
## 5 10x10m     6.25
```

- 3.) In at least a paragraph, answer which is the most precise (least variable) plot size and why.
- 4.) In at least a paragraph, answer which is the most efficient plot size and why.
- 5.) Converting the density to frequency, what plot size worked well for frequency?
- 6.) Theoretically frequency (p in %) and density (d) are related. Some suggest that the conversion from frequency to density is $d = -\ln(1-p/100)$. Calculate the estimated density for the 5x5m quadrat frequency data. (Convert to plants per m^2) Does the answer come out the same or different from the actual density answer?
- 7.) Write at least a paragraph commenting about this exercise.
- 8.) For the 2.5x2.5m plot size, calculate the sample size to produce, with 95% probability, a sample mean within 10% of the true population mean. What does this say about our ability to detect changes with the sample size of 20?